



Barker
College

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Student Number

Year 12 HSC Mathematics Advanced
Assessment Task 1 2024
(35 Marks)

Sequences and Series
Series and Finance
Discrete Probability Distributions
Graphs and Equations

Staff Involved:

JAI GPF AXD

RAS* DOB PCB ECB EJK SJB

8:30am

Wednesday, 14th February

140 copies

General

Instructions:

- Working time - 50 minutes
- Write using black pen
- Write your student number on all pages
- NESA approved calculators may be used
- A reference sheet is provided separately
- Show relevant mathematical reasoning and/or calculations
- Marks may not be awarded for careless or poorly arranged working

Section I - 5 marks

- Attempt Questions 1 - 5
- Circle the most correct answer

Section II- 30 marks

- Attempt Questions 6 - 15
- Write your answers in the spaces provided

Section I

Questions 1 to 5 are multiple choice questions.

Circle the most correct answer.

1. The first three terms in a sequence are 20, 16 and 12.
What is the 11th term in this sequence?
- A. 60
- B. -20
- C. -24
- D. -20 971 520

2. The probability distribution of a random variable X is shown below:

x	0	1	2	3	4
$P(X=x)$	$\frac{3}{10}$	$\frac{1}{20}$	k	$\frac{1}{20}$	$2k$

What is the value of k ?

- A. $\frac{2}{15}$
- B. $\frac{1}{5}$
- C. $\frac{1}{10}$
- D. $\frac{2}{5}$

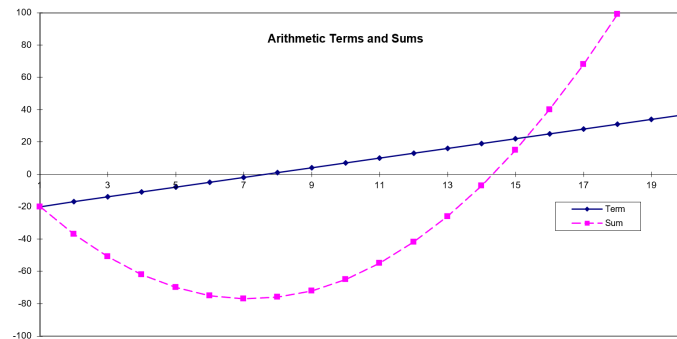
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3. An infinite geometric series has a first term of 8 and a limiting sum of 24.
What is the common ratio?

- A. $\frac{1}{3}$
B. $\frac{1}{2}$
C. $\frac{2}{3}$
D. $\frac{24}{16}$

4. The following graph was produced from **ArithmeticSeries.xls** from the Investigation.



Which of the following statements is true about the graph?

- A. $a < 0$ and $d > 0$
B. $a > 0$ and $d > 0$
C. $a > 0$ and $d < 0$
D. $a < 0$ and $d < 0$

5. A function $y = f(x)$ was vertically dilated by a factor of 2, translated 4 units to the right then horizontally dilated by a factor of 3.

Which of the following is the equation of the new function?

- A. $y = 2f\left(\frac{x}{3} + 4\right)$
B. $y = 3f\left(\frac{x}{2} - 4\right)$
C. $y = 3f(2x + 4)$
D. $y = 2f\left(\frac{x}{3} - 4\right)$

End of Section I

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Section II

Questions 6 to 15 are extended response questions.

Answer in the spaces provided.

6. Is 309 a term in the sequence 13, 19, 25, 31, ... ?
Justify your answer with mathematical working. 2

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7. 3, x , 108 are the first three terms of a geometric sequence. Find x . 2

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8. By first rewriting as an expression including a series, convert $1.\overline{34}$ to an improper fraction. 2

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9. Lionel and Sam are competing to fill the final place in a mixed soccer team.

Lionel created a discrete probability distribution where the random variable X is the number of goals he scores in a game.

x	0	1	2	3
$P(X = x)$	0.35	0.15	0.1	0.4

- i. Calculate $E(X)$ of this distribution. 1

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- ii. Calculate $\text{Var}(X)$, showing working. 2

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Sam did similar calculations and found she had an expected value of 1.03, with a variance of 0.3.

- iii. With reference to the above calculations, who should fill the position in the team? Justify your answer. 2

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10. A function $y = f(x)$ was transformed to $y = 2f\left(\frac{x}{3}\right) - 4$. If the point $P(4, 5)$ was on the original function, what are the new coordinates of the image of P ? **2**

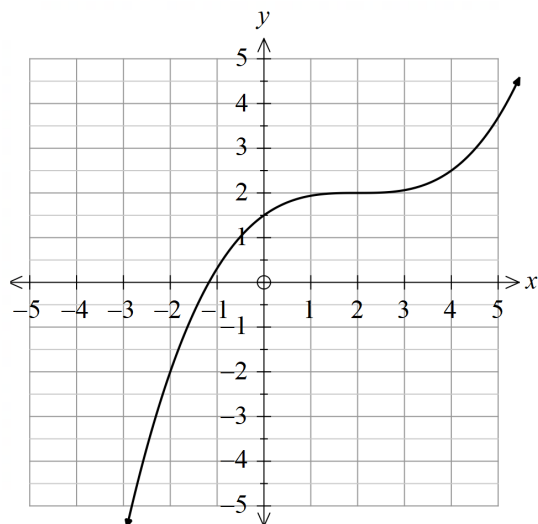
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11. Consider the graph of $y = f(x)$ as shown.



On the above number plane, draw a sketch of the function $y = 2f(x) - 4$. Ensure that you show x and y intercepts.

3

12. An underground dice game is played at school and has the following rules.

- An entry fee of \$1 is paid and a dice is rolled.
- If you roll a 1, you receive nothing.
- If you roll a prime number, you are paid \$2.
- If you roll a composite number, you pay an extra \$1.

Let x be the net financial outcome from the game, including the entry fee.

Complete the following probability distribution table for this game.

x			
$P(X = x)$			

Find the expected value of the game and determine whether you would expect to win or lose money playing it. **3**

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13. Given the condition for a limiting sum to occur is when $|r| < 1$, use the formula for the sum of a geometric series, $S_n = \frac{a(1-r^n)}{1-r}$, to derive the formula for a limiting sum, $S_\infty = \frac{a}{1-r}$

Ensure that you clearly explain your working.

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14. The graph of the function $f(x) = x^2$ is translated p units to the left then dilated vertically by a factor of q .

The equation of the transformed function is now $g(x) = 3x^2 + 24x + 48$.

Find the values of p and q .

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15. William is saving for a holiday. On the 1st January 2024, he deposited \$4000 into an account earning 2% per month. On the first of each month after this, he deposited \$2000 into the same account, making his final deposit on 1st January, 2025.

- i. By first building a series, show that the amount in William's account on 1st January, 2025 after the final deposit was made, is \$31 897.15. 3

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Question 15 continues on the next page

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- ii. William decides to leave this \$31 897.15 in the same account, earning 2% interest per month. On 1st February, 2025 and on every first of the month afterwards, he withdraws \$3000 from the account.

By first building a series, determine how many withdrawals of \$3000 William will be able to make. **3**

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End of Assessment

1 $a = 20, d = -4$
 $T_n = a + (n-1)d$
 $T_{11} = 20 + (11-1)(-4)$
 $= -20$ [B]

2 $\frac{3}{10} + \frac{1}{20} + k + \frac{1}{20} + 2k = 1$
 $3k + \frac{2}{5} = 1$
 $3k = \frac{3}{5}$
 $k = \frac{1}{5}$ [B]

3 $a = 8, S_{\infty} = 24$
 $S_{\infty} = \frac{a}{1-r}$
 $24 = \frac{8}{1-r}$
 $24(1-r) = 8$
 $1-r = \frac{8}{24}$
 $r = 1 - \frac{8}{24}$
 $r = \frac{2}{3}$ [C]

4 The term line starts below the x-axis, $\therefore a < 0$
 The sum of the terms begins to increase once the term becomes positive, $\therefore d > 0$. [A]

- 5 ① dilated vert. by a factor of 2 $\Rightarrow$ replace $y \rightarrow \frac{y}{2}$
 ② translated 4 units right $\Rightarrow$ replace $x \rightarrow x-4$
 ③ dilated horiz. by a factor of 3 $\Rightarrow$ replace $x \rightarrow \frac{x}{3}$

① $y = f(x) \Rightarrow \frac{y}{2} = f(x)$
 $y = 2f(x)$
 ② $y = 2f(x) \Rightarrow y = 2f(x-4)$
 ③ $y = 2f(x-4) \Rightarrow y = 2f(\frac{x}{3}-4)$
 [D]

6 $a = 13, d = 6$
 $T_n = 13 + (n-1)(6)$
 $= 13 + 6n - 6$
 $= 6n + 7$

Let $T_n = 309$
 $309 = 6n + 7$
 $302 = 6n$
 $n = \frac{151}{3}$
 $\therefore 309$ is NOT a term in the sequence because n is not an integer.

7 $\frac{T_2}{T_1} = \frac{T_3}{T_2}$
 $\frac{x}{3} = \frac{108}{x}$
 $x^2 = 324$
 $x = \pm 18$

8 $1.\dot{3}\dot{4} = 1 + 0.34 + 0.0034 + \dots$
 $a = 0.34, r = 0.01$
 $= 1 + \frac{0.34}{1-0.01}$
 $= 1 + \frac{34}{99}$
 $= \frac{133}{99}$

9 (i) $E(X) = 0(0.35) + 1(0.15) + 2(0.1) + 3(0.4)$
 $= 1.55$

(ii) $E(X^2) = 0^2(0.35) + 1^2(0.15) + 2^2(0.1) + 3^2(0.4)$
 $= 4.15$

$\text{Var}(X) = E(X^2) - [E(X)]^2$
 $= 4.15 - 1.55^2$
 $= 1.75$

(iii)	Lionel	Sam
$E(X)$	1.55	1.03
$\text{Var}(X)$	1.75	0.3

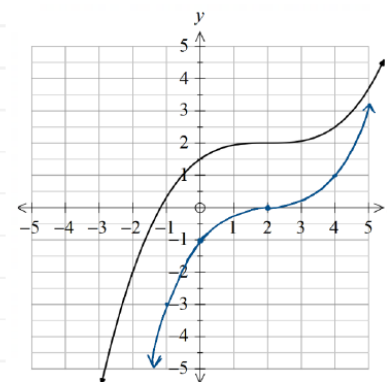
Answer could be either:

① Lionel $\rightarrow$ Because even though his variance is larger, meaning he is less consistent than Sam, he is expected to score more goals than Sam each game.

② Sam $\rightarrow$ Although Sam is expected to score fewer goals than Lionel, he is a more consistent player.

10 $y = 2f(\frac{x}{3}) - 4$
 Horizontally:
 * dilated by a factor of 3
 * no translation
 Vertically:
 * dilated by a factor of 2
 * translated down 4 units
 $(4, 5) \rightarrow (4 \times 3, 5 \times 2 - 4) = (12, 6)$

11 $y = 2f(x) - 4$
 Horizontally:
 * no dilation or translation
 Vertically:
 * dilated by a factor of 2
 * translated down 4 units
 $(2, 2) \rightarrow (2, 0) \Rightarrow$ new x-int.
 $(0, 1.5) \rightarrow (0, -1) \Rightarrow$ new y-int.
 $(4, 2.5) \rightarrow (4, 1)$



12

x	-2	-1	1
$P(X=x)$	$\frac{2}{6}$	$\frac{1}{6}$	$\frac{3}{6}$

$$E(X) = -2\left(\frac{2}{6}\right) - 1\left(\frac{1}{6}\right) + 1\left(\frac{3}{6}\right)$$

$$= -\frac{1}{3}$$

$\therefore$ It is expected that around 33 cents will be lost per game, taking into account the \$1 entry fee.

13

$$S_{\infty} = \frac{a(1-r^{\infty})}{1-r}$$

Because $|r| < 1$, $r^{\infty} \rightarrow 0$

$$S_{\infty} = \frac{a(1-0)}{1-r}$$

$$= \frac{a}{1-r}$$

14

Method 1

$$g(x) = 3x^2 + 24x + 48$$

$$= 3(x^2 + 8x + 16)$$

$$= 3(x+4)^2$$

In the form $y = k f(a(x+b)) + c$

$\therefore p = 4$ and $q = 3$

Method 2

$$g(x) = q(x+q)^2$$

$$= q(x^2 + 2px + p^2)$$

$$= qx^2 + 2pqx + p^2q$$

$$qx^2 + 2pqx + p^2q = 3x^2 + 24x + 48$$

Equating coefficients:

$$q = 3 \text{ and } 2pq = 24$$

$$2p(3) = 24$$

$$6p = 24$$

$$p = 4$$

15

(i) $A_1 = 4000(1.02) + 2000$

$$A_2 = A_1 \times 1.02 + 2000$$

$$= [4000(1.02) + 2000](1.02) + 2000$$

$$= 4000(1.02)^2 + 2000(1.02) + 2000$$

$$A_3 = A_2 \times 1.02 + 2000$$

$$= [4000(1.02)^2 + 2000(1.02) + 2000](1.02) + 2000$$

$$= 4000(1.02)^3 + 2000(1.02)^2 + 2000(1.02) + 2000$$

$\vdots$

$$A_{12} = 4000(1.02)^{12} + 2000(1.02)^{11} + 2000(1.02)^{10} + \dots + 2000$$

$$= 4000(1.02)^{12} + 2000(1.02)^{11} + 2000(1.02)^{10} + \dots + 2000$$

$$= 4000(1.02)^{12} + 2000 \left(\frac{1.02^{12} - 1}{0.02} \right)$$

$$= \$31897.15$$

(ii) A_1 now represents the 1st Feb 2025

$$A_1 = 31897.15(1.02) - 3000$$

$$A_2 = A_1 \times 1.02 - 3000$$

$$= [31897.15(1.02) - 3000](1.02) - 3000$$

$$= 31897.15(1.02)^2 - 3000(1.02) - 3000$$

$$A_3 = A_2 \times 1.02 - 3000$$

$$= [31897.15(1.02)^2 - 3000(1.02) - 3000](1.02) - 3000$$

$$= 31897.15(1.02)^3 - 3000(1.02)^2 - 3000(1.02) - 3000$$

$\vdots$

$\vdots$

$$A_n = 31897.15(1.02)^n - 3000(1.02)^{n-1} - 3000(1.02)^{n-2} - \dots - 3000$$

$$= 31897.15(1.02)^n - 3000(1.02^{n-1} + 1.02^{n-2} + \dots + 1)$$

$$= 31897.15(1.02)^n - 3000 \left(\frac{1.02^n - 1}{0.02} \right)$$

$$= 31897.15(1.02)^n - \frac{3000}{0.02}(1.02^n - 1)$$

$$= 31897.15(1.02)^n - 150000(1.02^n - 1)$$

$$= 31897.15(1.02)^n - 150000(1.02)^n + 150000$$

$$= -118102.85(1.02)^n + 150000$$

When $A_n = 0$

$$0 = -118102.85(1.02)^n + 150000$$

$$118102.85(1.02)^n = 150000$$

$$(1.02)^n = \frac{150000}{118102.85}$$

$$n = \log_{1.02} \left(\frac{150000}{118102.85} \right)$$

$$= 12.07 \text{ (to 2dp)}$$

$\therefore$ William can make 12 withdrawals.